SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models**.** *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4 **Total Marks:40**

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:
I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: *E.Ragul*

1.ANSWER

1. Book Bank Process

Question Explanation

A college has a **Book Bank System** where students can borrow books.

The process is:

1. Student submits request.
2. Administrator enters student details.
3. Central Computer verifies the student.
4. Approval is sent to the office.
5. Office issues books to the student.

The question asks you to develop UML diagrams for this process.

Actors

- Student
- Administrator
- Central Computer
- Office

Use Cases

- Register Student
- Submit Book Request
- Enter Student Details
- Verify Student Details
- Approve Request
- Issue Books

Student

Attributes

- StudentID
- Name
- Department
- Year

Methods

- requestBook()
- viewStatus()

Attributes

- AdminID
- Name

Methods

- enterDetails()
- sendForVerification()

Central Computer

Attributes

- Database

Methods

- verifyStudent()
- approveRequest()

Office

Attributes

- OfficeID

Methods

- issueBook()

Book

Attributes

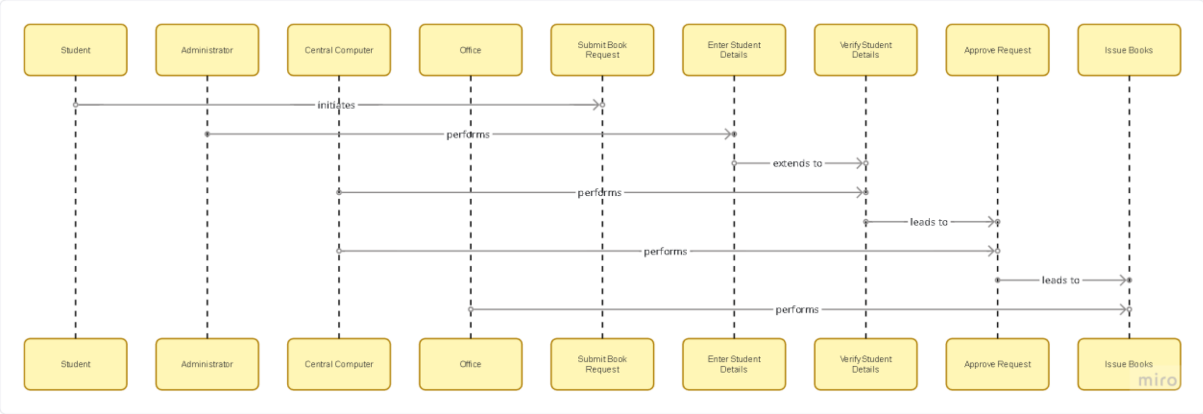
- BookID
- Title
- Author
- Availability

Methods

- issueBook()
- returnBook()

Relationships

- Student → requests → Book
- Administrator → enters → Student Details
- Central Computer → verifies → Student
- Office → issues → Book



2.ANSWER

Exam Registration Portal

A student wants to register for an examination.

The process is:

1. Candidate submits details.
2. Administrator enters details.
3. Central Computer verifies.
4. Approval is given.
5. Office issues Hall Ticket.

Actors

- Candidate
- Administrator
- Central Computer
- Office

Use Cases

- Register Candidate
- Enter Candidate Details
- Verify Details
- Approve Registration
- Generate Hall Ticket
- Issue Hall Ticket

Classes

Candidate

Attributes

- CandidateID

- Name
- Department
- Semester

Methods

- registerExam()
- downloadHallTicket()

Administrator

Attributes

- AdminID
- Name

Methods

- enterDetails()
- forwardRequest()

CentralComputer

Attributes

- Candidate Database

Methods

- verifyCandidate()
- approveRegistration()

Office

Attributes

- OfficeID

Methods

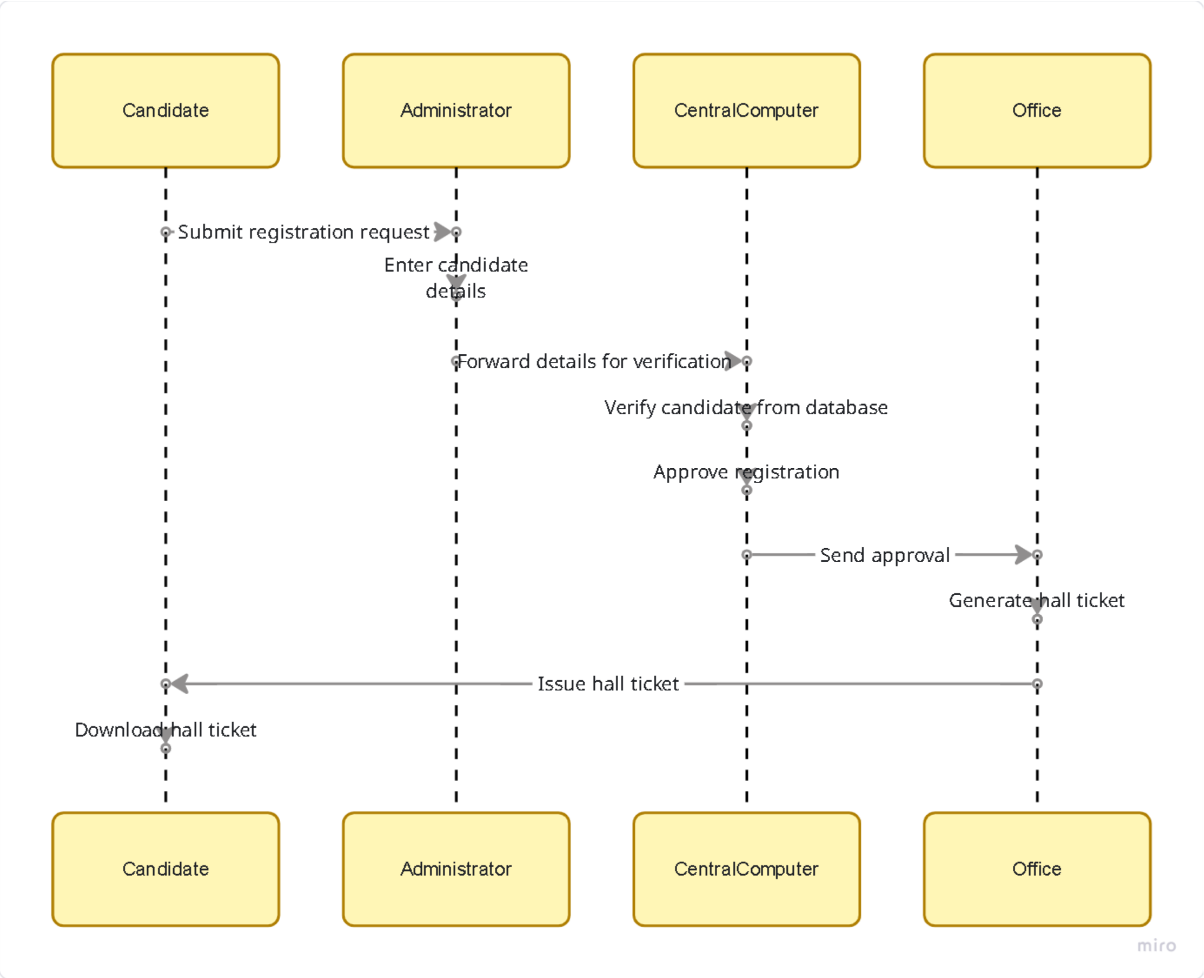
- generateHallTicket()
- issueHallTicket()

Relationships

- Candidate → registers
- Administrator → submits details
- Central Computer → verifies
- Office → issues Hall Ticket

Sequence Flow

1. Candidate registers.
2. Administrator enters details.
3. Central Computer verifies.
4. Approval sent.
5. Office issues Hall Ticket.



3.ANSWER

A customer wants to purchase products.

The process is:

1. Customer places an order.
2. Stock Dealer receives the order.
3. Central Stock System verifies stock.
4. Items are delivered.
5. Stock quantity is updated.

Actors

- Customer
- Stock Dealer

- Central Stock System

Use Cases

- Place Order
- Verify Stock
- Process Order
- Deliver Items
- Update Stock

Classes

Customer

Attributes

- CustomerID
- Name
- Address

Methods

- placeOrder()
- makePayment()

StockDealer

Attributes

- DealerID
- Name

Methods

- processOrder()
- deliverItems()

Stock

Attributes

- ItemID
- ItemName
- Quantity
- Price

Methods

- checkAvailability()

- updateStock()

Order

Attributes

- OrderID
- OrderDate
- Quantity

Methods

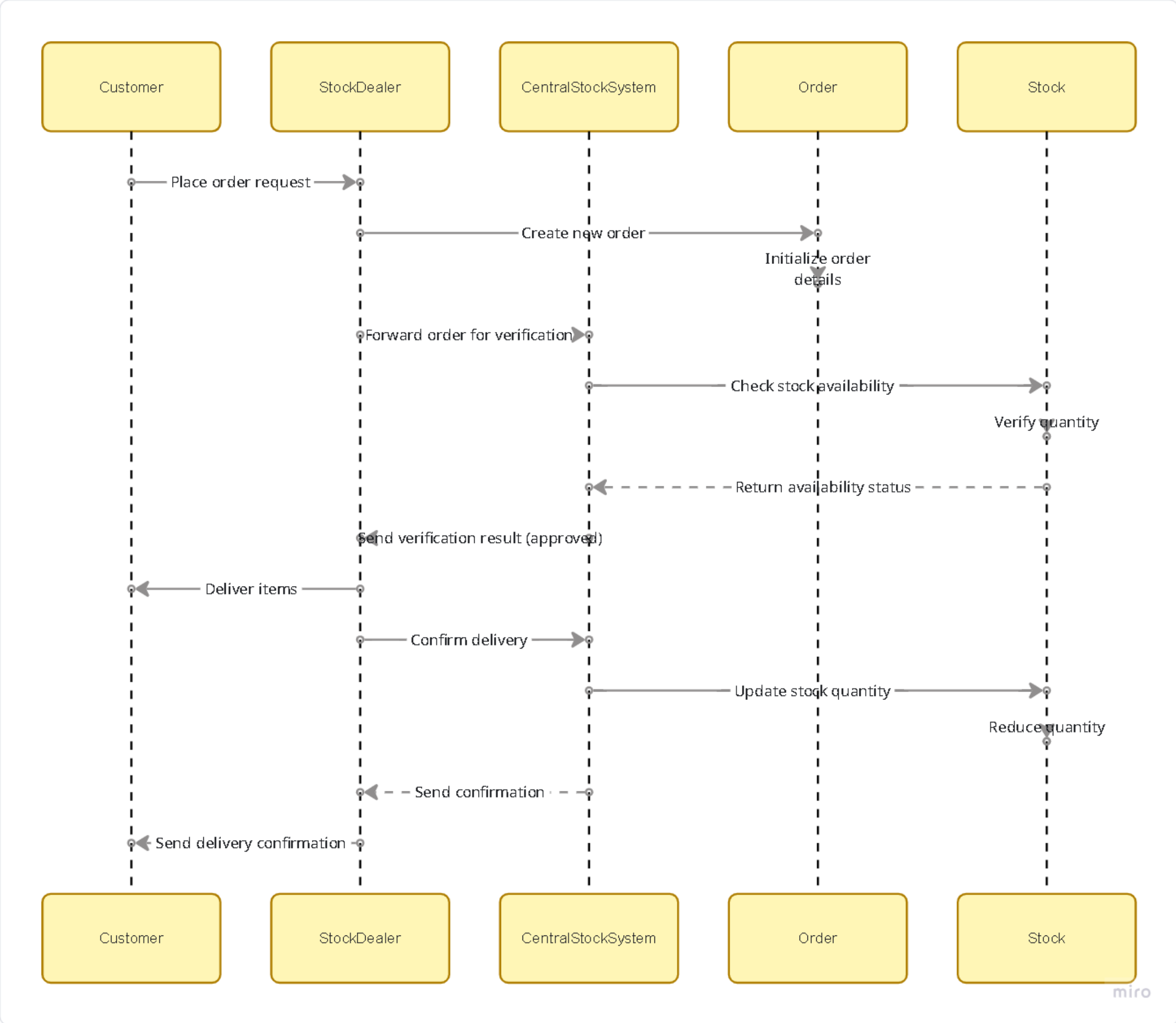
- createOrder()
- cancelOrder()

Relationships

- Customer → places → Order
- Order → contains → Stock Items
- Dealer → processes → Order
- Stock System → updates → Stock

Sequence Flow

1. Customer places order.
2. Dealer forwards order.
3. Stock system verifies stock.
4. Items delivered.
5. Stock updated.



4.ANSWER

Question Explanation

A student wants to reserve a course online.

The process is:

1. Student browses courses.
2. Student selects a course.
3. University checks seat availability.
4. If seats are available, the student is enrolled.

The question also asks you to **identify classes, attributes, relationships, and develop a use case model**.

Actors

- Student
- University
- Central Management System

Use Cases

- Login
- Browse Courses
- Select Course
- Check Seat Availability
- Enroll Student
- View Enrollment Status

Classes

Student

Attributes

- StudentID
- Name
- Department

Methods

- browseCourse()
- reserveCourse()

Attributes

- CourseID
- CourseName
- Duration
- AvailableSeats

Methods

- checkAvailability()
- reserveSeat()

University

Attributes

- UniversityID
- Name

Methods

- approveEnrollment()
- manageCourses()

ManagementSystem

Attributes

- Database

Methods

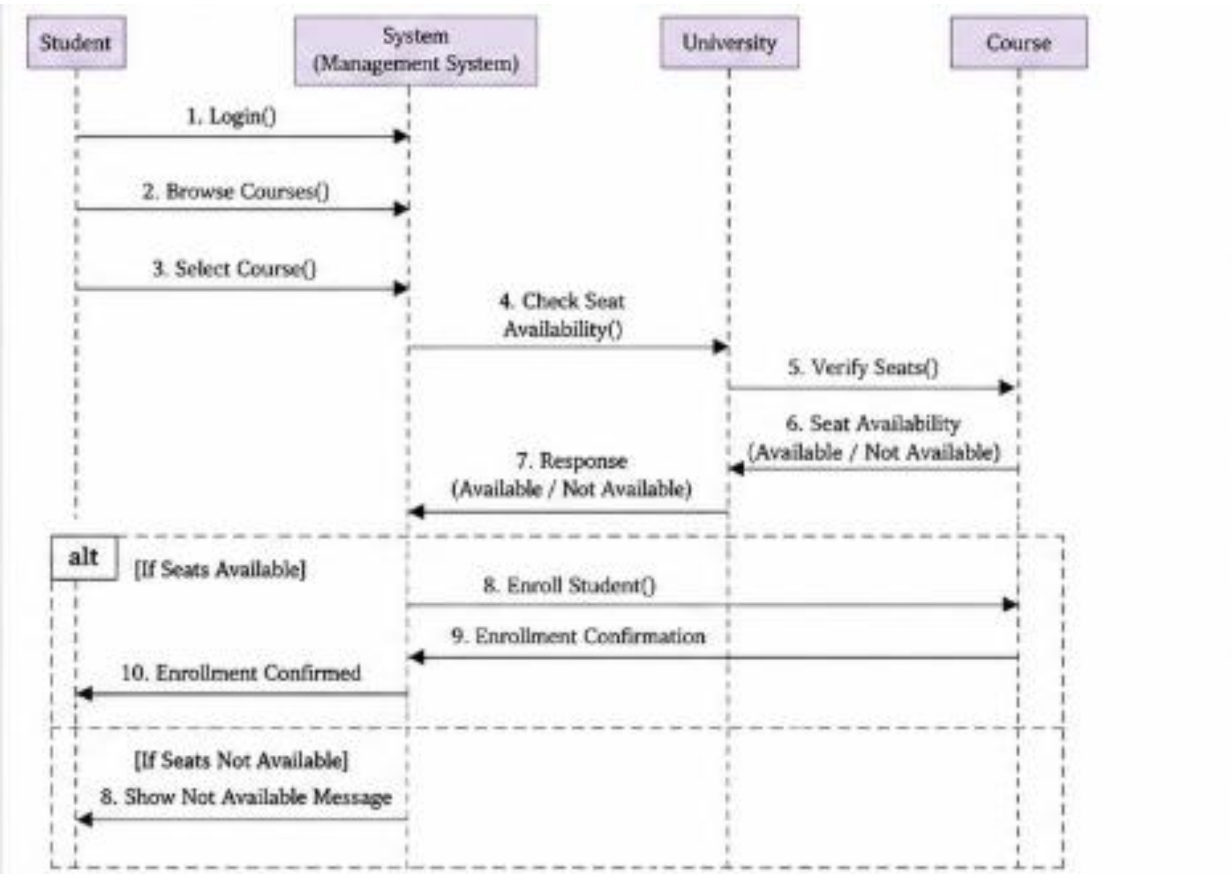
- verifySeats()
- enrollStudent()

Relationships

- Student → selects → Course
- University → offers → Course
- Management System → manages → Course
- Student → enrolled in → Course

Sequence Flow

1. Student logs in.
2. Browses courses.
3. Selects course.
4. System checks seats.
5. If available, enrollment is confirmed.



RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation